BTS-IV(S)-04.21-0253 Reg. No. [| | | | | | |]

A

B.Tech. Degree IV Semester Supplementary Examination April 2021

SE 15-1402 HEAT AND MASS TRANSFER OPERATIONS

(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) State the Fourier's Law of heat conduction.

(b) What is meant by convection? Enlist the different types of convection.

(c) Explain the significance of log mean temperature difference.

(d) Discuss the different types of heat exchangers.

(e) Compare between gray and black body.

(f) What is diffusion? State the Fick's law of diffusion.

(g) Differentiate between equilibrium line and operating line in absorption calculations.

(h) What is reflux ratio? What is its significance?

(i) Explain the principle of leaching and equipment used for the same.

(j) Compare between azeotropic and extractive distillation.

PART B

(4 × 10 = 40)

II. (a) Derive an expression for the steady state heat conduction through a plane wall. (4)

(b) A furnace walls made up of three layers, one of fire brick, one of insulating brick and one of red brick. The inner and outer surfaces are at 870°C and 40°C respectively. The respective co- efficient of thermal conduciveness of the layer are 1.0, 0.12 and 0.75 W/mK and thicknesses are 22 cm, 7.5 cm and 11 cm. assuming close bonding of the layer at their interfaces, find the rate of heat loss per sq.meter per hour and the interface temperatures. (6)

OR

III. (a) How heat is transferred from condensing vapors? (3)

(b) With the help of neat diagram, explain the steps involved in the process of boiling. (7)

IV. Water flows at the rate of 65 kg/min through a double pipe counter flow heat exchanger. Water is heated from 50°C to75°C by an oil flowing through the tube. The specific heat of the oil is 1.780 kJ/kg.K. The oil enters at 115°C and leaves at 70°C.the overall heat transfer co-efficient is 340 W/m^{2}K. Calculate the following: (10)

(i) Heat exchanger area
(ii) Rate of heat transfer.

OR

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V. (a) Two large plates are maintained at a temperature of 900 K and 500 K respectively. Each plate has area of $6m^2$. Compare the net heat exchange between the plates when both plates are black. (5)

(b) A single effect evaporator is used to concentrate 9070 kg/hr of 20% caustic soda solution to 50% solids. The gauge pressure of steam is 1.37atm. The absolute pressure in the vapor space is 100 mm Hg. The overall heat transfer coefficient is estimated to be 1400 $W/m^2 °C$ and the feed temperature is 37.8 °C. Calculate the amount of steam consumed. (5)

Data: Enthalpy of feed at 37.8 °C = 127.9245 kJ/kg
Enthalpy of thick liquor = 514.0239 kJ/kg
Enthalpy of vapour = 2672.46 kJ/kg
Heat of vaporization of steam at 1.37 atm = 2184.0201 KJ/ Kg
Condensation temperature of steam = 126.11°C

VI. (a) What are the major factors considered while choosing the solvent for absorption? (4)

(b) Explain the different theories of mass transfer. (6)

OR

VII. An air-acetone mixture containing 9% acetone by volume is to be washed with water in a counter-current plate tower to recover 90% acetone. The air rate will be 1000 m^3 of pure air per hour at standard conditions and the water rate will be 3500 kg/hr. The tower will operate at 20°C and 1 atm pressure. Under the operating condition, the equilibrium relation may be expressed as $Y^* = 1.68X$, where X and Y are moles of acetone per mole of acetone free water and air respectively. Determine graphically the number of theoretical plates required for the purpose. (10)

VIII. A rectification column is fed with an equimolal mixture of pentane and hexane. The feed is 50% vaporised. The top product should contain 90 mol% pentane and the bottom product should contain 90 mol% hexane. Determine the actual reflux ratio which is 2 times the minimum reflux ration and the number of theoretical plates for the operation. (10)

OR

IX. (a) With the help of a neat diagram, explain the drying curve and the stages involved in drying. (4)

(b) Explain with the help of neat diagram the construction and working of rotary dryer and spray dryer. (6)

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